DAET TOURIST INFORMATION, ENGAGEMENT, AND MANAGEMENT SYSTEM WITH REAL-TIME UPDATES

(Connect Daet)

A Capstone Project Presented to the Faculty of [College / Department]

[Name of School]

In Partial Fulfillment of the Requirements for the Degree

[Bachelor of Science in Information Technology]

[Proponent 1]

[Proponent 2]

[Proponent 3]

[Month Year]

APPROVAL SHEET

[Insert the approval sheet from Annex J with the names and signatures of the adviser, panel members, and dean.]

DECLARATION OF AI USE

[Insert the declaration from Annex J describing which AI tools were used, for what tasks (e.g., drafting, code assistance, editing), and how outputs were reviewed and verified by the proponents.]

CERTIFICATION OF ADVISER

[Insert the adviser's certification from Annex J.]

CERTIFICATION OF LANGUAGE EDITOR

[Insert the language editor's certification from Annex J.]

CERTIFICATION OF LAYOUT AND FORMATTING EDITOR

[Insert the layout and formatting editor's certification from Annex J.]

RECOMMENDATION OF THE DEAN

[Insert the dean's recommendation from Annex J.]

EXECUTIVE SUMMARY

The Daet Tourist Information, Engagement, and Management System with Real-Time Updates (Connect Daet) is a web-based Progressive Web Application developed for the Daet Municipal Tourism Office to address fragmented tourism information, untracked visitor inquiries, and the absence of an official real-time advisory channel. The system provides a searchable directory of attractions, accommodations, restaurants, and local businesses with maps and directions; category-targeted real-time alerts delivered to in-app inboxes and devices, with scheduling and reach tracking; pre-moderated reviews, blogs, and forums; tracked inquiries; and a role-based administrative console for Administrators, Tourism Officers, and Moderators with audit logs and exportable analytics.

The study used a descriptive-developmental design and the Agile Scrum methodology. The system was built with Next.js, React, Tailwind CSS, and Supabase (PostgreSQL, Authentication, Storage, Realtime, and Edge Functions). It passed 160 automated tests and a security review based on the OWASP Top 10. [Summarize the Usability evaluation: the overall weighted mean on the five-point Likert scale, its verbal interpretation, and the key recommendation.]

ACKNOWLEDGEMENT

[Write your acknowledgement.]

DEDICATION

[Write your dedication.]

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CHAPTER 1

INTRODUCTION

This chapter presents the foundational framework of the study. It describes the system being developed, the conditions that made it necessary, the organization it serves, the problems it addresses, the objectives it pursues, and the boundaries within which it operates.

Project Context

The Daet Tourist Information, Engagement, and Management System with Real-Time Updates (hereafter referred to as **Connect Daet**) is a web-based Progressive Web Application (PWA) that serves as a single official online portal for tourism in the Municipality of Daet, Camarines Norte. It replaces the current patchwork of social media posts, static pages, and unofficial blogs with one source of verified tourism information that is maintained directly by the Daet Municipal Tourism Office.

The project builds three connected components. The first is a **public tourism portal** where visitors can browse and search attractions, accommodations, restaurants, local businesses, and events, view photos, maps, operating hours, entrance fees, and contact details, and get directions to each place. The second is a **real-time alert and engagement layer** that delivers official advisories (weather, sea and beach conditions, road closures, power and water interruptions, health, and event changes) to subscribers through in-app notifications and browser push notifications, and that lets registered tourists write reviews, publish travel blogs, join discussion forums, save places to a trip plan, and send inquiries and feedback to the tourism office. The third is a **role-based administrative console** where tourism personnel manage listings, events, alerts, community content, and accounts, and where they view analytics that can be exported as PDF and Excel reports.

The intended users are grouped into five roles. **Visitors** (guests without an account) can read all public information and alerts. **Registered tourists** can additionally interact with content and receive personalized alerts. **Tourism Officers** maintain listings, events, alerts, reviews, inquiries, and tourism analytics. **Moderators** review community submissions, handle reports, warnings, posting bans, suspensions, and appeals, and

maintain forum categories and community guidelines. **Administrators** manage staff and tourist accounts, roles and permissions, system settings, security policies, audit logs, and database backups.

At a macro level, users access the system through any modern web browser on a phone, tablet, or computer. The application is built with Next.js and React and is hosted on a cloud platform; it stores data in a PostgreSQL database managed through Supabase, which also provides authentication, file storage, and real-time change notifications. When a Tourism Officer publishes an alert, the system writes the alert to every subscriber's notification inbox in a single database operation and requests a push notification to their devices; open pages refresh automatically when tourism content changes. Every privileged action is checked against the user's role on the server and recorded in an audit log.

Background of the Study

Tourism is one of the most information-intensive industries. Travelers decide where to go, where to stay, and what to do largely on the basis of information found online, and destinations increasingly compete on the quality and timeliness of the information they provide (Buhalis & Law, 2008). Recent reviews confirm that this remains central: a 2023 systematic literature review of smart tourism destinations found that research increasingly links destination technology to sustainable development and to better management of visitor information (El Archi et al., 2023). The concept of *smart tourism* extends this idea: destinations are expected to use information and communication technologies to deliver accurate, real-time, and personalized information to visitors while giving tourism managers data to guide decisions (Gretzel et al., 2015).

In the Philippines, tourism is recognized by law as an engine of national development. Republic Act No. 9593, the Tourism Act of 2009, declares tourism an indispensable element of the national economy and directs local government units to develop and promote their tourism destinations. The National Tourism Development Plan 2023–2028 of the Department of Tourism likewise emphasizes digitalization and the improvement of tourist information and services (Department of Tourism, 2023). International arrivals have recovered steadily since the pandemic: the Department of Tourism recorded 5,949,350 visitor arrivals in 2024, of whom 5,438,697 were foreign

tourists (Manila Bulletin, 2025), and 6,484,060 arrivals in 2025, a 0.76 percent increase that remained below the pre-pandemic level of about eight million (DZRH News, 2026). Because the country is also highly exposed to typhoons and other hazards, timely public advisories are a recognized government responsibility under Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010) and Republic Act No. 10639 (Free Mobile Disaster Alerts Act).

Daet, the capital municipality of Camarines Norte, is steadily emerging as a key tourist destination in the Bicol Region. Its reputation is anchored by Bagasbas Beach, a premier spot for surfers and beach enthusiasts, and it is further enriched by cultural celebrations such as the Pinyasan Festival, which honors the town's thriving pineapple industry. In 2024, Camarines Norte recorded 888,810 domestic tourist arrivals, the third highest among the six Bicol provinces after Albay and Camarines Sur, but only 474 foreign arrivals (Bicol Mail, 2025), and the provincial economy grew by 3.7 percent in 2023 (Philippine Statistics Authority, 2024). The very small number of foreign visitors suggests that the province is not yet easy to discover and plan for online. [Add Daet-specific arrival figures for 2022–2025 from the Daet Municipal Tourism Office, if available.] This combination of natural and cultural assets has led to a growing number of visitors each year, making tourism a vital part of the local economy.

However, the municipality's capacity to manage and disseminate tourism information has not kept pace with this growth. Tourists are often left to navigate static websites, unofficial blogs, and scattered social media pages to plan their visits. The Daet Municipal Tourism Office currently depends almost exclusively on its Facebook page as its digital communication channel. This reliance on a single, general-purpose social media platform presents several operational problems.

Foremost among these is **information fragmentation and inconsistency**. Because there is no single source of truth, tourists frequently encounter conflicting details about operating hours, event schedules, and available amenities across different platforms, which erodes trust and causes confusion. Compounding this is **outdated content**: static websites are rarely updated, and information on social media is quickly buried under newer posts. The Facebook page is also ill-equipped to handle a growing volume of **visitor inquiries**;

questions posted in comments or private messages can be overlooked, left unanswered, or require staff to type the same answers repeatedly. When sudden changes occur, such as weather-related closures, road works, or last-minute changes to festival schedules, there is **no mechanism for immediate, official, and targeted alerts**, which can directly affect tourist safety and satisfaction.

For tourism personnel, managing information is a manual and reactive task. They lack a centralized dashboard for updating attraction details, posting events, or controlling what is presented to the public. There is no structured way to collect, analyze, and act on tourist feedback and ratings, so insight into visitor satisfaction, popular attractions, and areas needing improvement is lost. Community discussions also take place on platforms the office does not control, with no moderation rules aligned to the municipality's standards. [Insert findings from your interviews or survey with the Daet Municipal Tourism Office and tourists that confirm these problems, e.g., number of unanswered inquiries, staff time spent, or tourist complaints.]

Existing general-purpose platforms only partly address these needs. Global review and mapping services provide listings and ratings but are not controlled by the local government and do not carry official advisories; social media pages lack structured listings, search, moderation tools, and analytics; and national government applications focus on national-level information rather than municipal tourism details. These gaps justify the development of a dedicated, integrated system that centralizes information, provides real-time official alerts, supports moderated community engagement, and gives the tourism office the data it needs to manage tourism proactively.

Company Profile

The client organization of this study is the **Daet Municipal Tourism Office**, an office of the Local Government Unit (LGU) of Daet, Camarines Norte. Under Republic Act No. 7160 (the Local Government Code of 1991) and Republic Act No. 9593 (the Tourism Act of 2009), local government units are responsible for developing, promoting, and regulating tourism within their jurisdiction. The office operates in the public administration and tourism sector.

The office is responsible for promoting Daet's attractions and events, coordinating festivals such as the Pinyasan Festival, maintaining information on local tourism establishments, assisting visitors and answering inquiries, and coordinating with other municipal offices and the provincial government on tourism programs. [Confirm and complete the office's actual functions with the Tourism Officer.]

Office address: [Address of the Daet Municipal Tourism Office]

Head of office: [Name and position of the Municipal Tourism Officer]

Mission: [Insert the office's official mission statement]

Vision: [Insert the office's official vision statement]

Organizational structure: [Describe each position shown in Figure 1.]

[Insert organizational chart showing the Municipal Mayor, the Municipal Tourism Officer, and the office staff]

Figure 1. Organizational Structure of the Daet Municipal Tourism Office

Significance of the Study

The development of Connect Daet is expected to benefit the following stakeholders, listed according to how directly they are affected by the system:

Daet Municipal Tourism Office (management and Tourism Officers). The system gives the office a single dashboard for managing listings, events, alerts, reviews, and inquiries, and provides analytics on traffic, popular places, ratings, and alert reach that support data-driven planning. It reduces repetitive manual work and enables proactive rather than reactive tourism management.

Tourists and Visitors. The system provides a reliable, centralized, and mobile-friendly source of accurate information about attractions, accommodations, restaurants, businesses, and events. Visitors receive official real-time alerts on safety and event changes, can plan trips, and can read reviews and community advice.

Moderators and System Administrators. Moderators gain structured tools for pre-moderating community content, handling reports, and enforcing community guidelines.

Administrators gain controls for accounts, roles, security policies, audit logs, and backups that protect the integrity of the platform.

Local Businesses and Establishments. Hotels, restaurants, and other tourism-related businesses gain visibility through the official places directory, along with verified reviews and the tourism office's public replies.

Local Government Unit of Daet. The system supports the municipality's tourism and e-governance goals, strengthens its destination image, and provides an official channel for public advisories.

Community and Residents. By supporting tourism growth, the system can contribute to local livelihood, and by showcasing local heritage and events, it can foster community pride.

Future Researchers and Developers. The documented design, database structure, security measures, and evaluation results can serve as a reference for similar municipal tourism and alert systems.

Statement of the Problem

Generally, this study addresses the problem that the Daet Municipal Tourism Office lacks a centralized, official, and real-time platform for disseminating tourism information, issuing advisories, engaging with visitors, and managing tourism data, which results in fragmented and outdated information, slow responses to inquiries, and decisions made without reliable data.

Specifically, it seeks to answer the following questions:

1. What are the operational challenges and communication gaps in the current tourism information dissemination and visitor engagement methods of the Municipality of Daet?
2. What technical, functional, and security requirements are needed in developing a real-time tourist information and management system for the Municipality of Daet?
3. How can the Daet Tourist Information, Engagement, and Management System with Real-Time Updates be designed and developed to provide: (a) a centralized and

mobile-responsive tourist information portal; (b) real-time alerts and announcements; (c) interactive features such as reviews, blogs, forums, and feedback; and (d) an administrative console with role-based access and analytics?

4. How acceptable is the developed system in terms of Usability based on the ISO/IEC 25010:2023 software quality model, as rated by the respondents on a five-point Likert scale, specifically in terms of appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness?

Objectives of the Study

The general objective of this study is to develop a functional, secure, and real-time system that enhances tourism information dissemination and stakeholder engagement for the Municipality of Daet.

Specifically, it aims to:

1. Identify the operational challenges and communication gaps in the current tourism information dissemination and visitor engagement methods of the Municipality of Daet.
2. Analyze the technical, functional, and security requirements of a real-time tourist portal, specifically the alert and push notification architecture, mobile-responsive standards, community moderation rules, and role-based access controls.
3. Design and develop the Daet Tourist Information, Engagement, and Management System with Real-Time Updates, integrating a mobile-responsive places and events portal, a real-time alert broadcast module, interactive community features (reviews, blogs, forums, feedback, and inquiries), and a role-based administrative console with analytics.
4. Evaluate the Usability of the system based on the ISO/IEC 25010:2023 software quality model using a five-point Likert scale questionnaire, in terms of appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness, with emphasis on mobile use by tourists.

Scope and Limitations

This study covers the design, development, and evaluation of Connect Daet, a web-based Progressive Web Application accessible through modern web browsers on desktop computers, tablets, and mobile phones. The system provides tourism information for the Municipality of Daet, Camarines Norte; issues real-time alerts; supports moderated community engagement; and supports tourism information management through a role-based administrative console. It uses PostgreSQL through Supabase as its database, Supabase Authentication for sign-in, Supabase Storage for photos and videos, and Supabase Realtime (WebSocket subscriptions) for live updates.

The system serves five user roles. Visitors can browse and search attractions, accommodations, restaurants, local businesses, and events; filter them by category, barangay, and rating; view photos, maps, contact details, and directions; read blogs, forum threads, and reviews; view current and past public alerts; read the About, Contact, and Community Guidelines pages; and subscribe a device to public alerts. Registered tourists can additionally register, sign in, recover passwords, edit profiles, save favorites and plan trip dates, write and edit reviews, publish blogs, take part in forums, react to and report content, choose alert categories, use a notification inbox, send inquiries and feedback and track their status, and earn points and badges.

Tourism Officers manage listings (with photos, map coordinates, operating hours, entrance fees, contact details, categories, tags, featured status, publishing, and archiving), events and festivals, real-time alerts (immediate or scheduled, targeted to all subscribers or to one category, edited, cancelled, or closed with an "all clear" message, with delivery and reach counts and reusable templates), public replies to reviews, inquiry responses, and tourism analytics with PDF and Excel export. Moderators manage pre-moderation queues for blogs, forum posts, and comments; reports; warnings, posting bans, temporary suspensions, and appeals; forum categories and threads; official announcements; and the community guidelines. Administrators manage staff and tourist accounts, roles and per-person permissions, system settings (site identity, logo, contact and municipality information, alert and place categories, password policy, session timeout, login lockout,

notification and email settings), audit logs, failed sign-in records, and database backup and restore.

The geographical coverage is limited to tourism destinations, establishments, and events within the Municipality of Daet. The development period covers [start month and year] to [end month and year].

The study is subject to the following limitations. The system is distributed only as a Progressive Web Application and is not published as a native application in the Google Play Store or Apple App Store. It does not include online booking, electronic payment, transport reservations, or tour package sales. The provision of server hardware, paid hosting plans, and domain registration is outside the scope; the system relies on cloud services (Supabase for the database, authentication, and storage, and a cloud host for the application), so storage, bandwidth, and connection limits depend on the plans used. Full functionality, including real-time updates and alerts, requires an internet connection; offline access is limited to static assets cached by the PWA.

Browser push notifications depend on the user granting permission and on browser support; on iPhones and iPads, the site must first be added to the Home Screen. Alerts are always delivered to the in-app inbox, but push delivery to a device is not guaranteed. Emails depend on the SendGrid service, maps and directions depend on Google Maps, and weather information on the administrator dashboard depends on the OpenWeather service. The accuracy and timeliness of information depend on the tourism office personnel who maintain it, and user-generated content is published only after moderator approval. The interface is available in English, Filipino, and Bikol for navigation and settings, but user-generated content is not automatically translated. The system does not offer sign-in through social media accounts. Compatibility is limited to current versions of modern browsers. Evaluation is limited to the Usability characteristic of ISO/IEC 25010, measured through a five-point Likert scale questionnaire; other quality characteristics such as performance efficiency, reliability, and security were verified through developer testing but were not rated by respondents. The evaluation does not measure its long-term effect on tourist arrivals or the local economy. Finally, the system is an information and advisory platform

and does not replace official emergency response channels; users are advised to contact local authorities during emergencies.

Definition of Terms

The following terms are defined operationally as they are used in this study:

Administrative Console. The restricted part of the system at /info/admin used by Administrators, Tourism Officers, and Moderators. Each role sees only the menus and actions its permissions allow.

Alert. An official advisory issued by a Tourism Officer with a title, message, category, severity (info, advisory, warning, or critical), and affected area, delivered to subscribers' inboxes and devices, either immediately or at a scheduled time.

All Clear. A follow-up message attached to a sent alert that tells subscribers the situation has ended.

Audit Log. A database record of every privileged action, such as role changes, deletions, sent alerts, and setting changes, including who performed it and when.

Engagement. The interactive features that let users take part in the platform: reviews, blogs, forums, reactions, reports, inquiries, and feedback.

Gamification. The use of points, badges, and rewards that registered tourists earn for approved participation.

Listing. A record in the places directory that describes an attraction, accommodation, restaurant, or local business.

Pre-moderation. The rule that blogs, forum posts, replies, and comments submitted by tourists are held as pending and become public only after a Moderator approves them.

Progressive Web Application (PWA). A website that can be installed on a device's home screen, caches static assets, and can receive push notifications, while still running in a web browser.

Push Notification. A message sent through the browser's push service to a subscribed device even when the site is not open.

Reach. The number of alert recipients who opened the alert in their inbox.

Real-Time Update. An automatic refresh of an open page shortly after the underlying data changes, using Supabase Realtime notifications, with a 30-second reconciliation fallback.

Role-Based Access Control (RBAC). The security rule set that grants each user role a fixed set of permissions, enforced on pages, on server APIs, and in the database.

Row-Level Security (RLS). Database policies in PostgreSQL that decide which rows each signed-in or anonymous user may read or change.

Supabase. The cloud platform that provides the system's PostgreSQL database, authentication, file storage, real-time notifications, and serverless functions.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter discusses the literature and studies, both local and foreign, gathered by the researchers to support the development of the proposed system. It also presents the similarities and differences between existing works and this study, and the theoretical and conceptual frameworks that guide it.

Related Literature

Information Technology in Tourism

Buhalis and Law (2008) reviewed two decades of research on information technology and tourism and concluded that the Internet fundamentally changed how travelers search for information and how destinations market themselves. Destinations that provide accurate, accessible, and current information online gain a competitive advantage, while those that rely on fragmented or outdated channels lose visibility. Xiang and Gretzel (2010) further showed that social media content dominates online travel searches, but that such content is user-generated, inconsistent, and not controlled by the destination. These findings support the need for an official, destination-managed information source rather than reliance on general social media pages.

Smart Tourism

Gretzel, Sigala, Xiang, and Koo (2015) defined smart tourism as tourism supported by integrated efforts at a destination to collect and use data from physical infrastructure, social connections, and government sources, in order to deliver real-time, personalized services to visitors and evidence for managers. Its key components are information delivered in real time, interaction between visitors and the destination, and data-driven management. Connect Daet applies these principles at the municipal level through real-time alerts, community interaction, and an analytics dashboard.

El Archi et al. (2023) reviewed 20 studies on smart tourism destinations and found that the field has moved from technology adoption toward the economic, social, and environmental effects of smart destinations, with emerging attention to the Sustainable

Development Goals and green information technology. The review supports the use of a destination-managed digital platform as part of sustainable tourism management.

Progressive Web Applications

Biørn-Hansen, Majchrzak, and Grønli (2017) examined Progressive Web Applications as a way to deliver app-like experiences through the web, including home-screen installation, offline caching through service workers, and push notifications, without maintaining separate native applications for each platform. This supports the choice of a PWA for a municipal office with limited development and maintenance resources.

Real-Time Communication on the Web

The WebSocket protocol (Fette & Melnikov, 2011) provides a persistent, two-way connection between a browser and a server, allowing servers to notify clients of changes without repeated polling. The Web Push protocol (Thomson, Damaggio, & Raymor, 2016) allows a server to deliver messages to a user's device through the browser vendor's push service even when the website is closed. Connect Daet uses WebSocket-based change notifications to refresh open pages and Web Push to deliver alerts to subscribed devices.

Role-Based Access Control and Web Application Security

Sandhu, Coyne, Feinstein, and Youman (1996) formalized Role-Based Access Control, in which permissions are attached to roles and users are assigned to roles, simplifying administration and enforcing least privilege. The OWASP Top 10 (OWASP Foundation, 2021) identifies broken access control, cryptographic failures, injection, and insecure design as the most critical web application risks. These works guided the system's five-role permission model, server-side authorization of every administrative API, database row-level security, input sanitization, and login lockout.

Gamification

Deterding, Dixon, Khaled, and Nacke (2011) defined gamification as the use of game design elements in non-game contexts. Points, badges, and rewards can encourage repeated, constructive participation. Connect Daet awards points and badges for approved contributions to encourage tourists to share reviews and stories.

Software Quality and Usability

ISO/IEC 25010:2023 (International Organization for Standardization, 2023) is the current product quality model. The 2023 revision defines nine quality characteristics, adding safety, and renames Usability as **Interaction Capability**: the degree to which a product can be interacted with by specified users to exchange information through the user interface to complete specific tasks in a variety of contexts of use. Its sub-characteristics are appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness; the earlier sub-characteristics of user interface aesthetics and accessibility were replaced by user engagement, inclusivity, and user assistance. This study uses the 2023 model as its evaluation basis. The Web Content Accessibility Guidelines 2.1 (World Wide Web Consortium, 2018) were followed as a design guide for inclusivity.

Legal Bases

Republic Act No. 9593 (Tourism Act of 2009) mandates local government units to develop and promote tourism. Republic Act No. 7160 (Local Government Code of 1991) devolves tourism promotion to LGUs. Republic Act No. 10173 (Data Privacy Act of 2012) requires that personal information be collected for a declared purpose, protected with reasonable security measures, and accessible to its owner; this guided the system's consent-based visit analytics, private account fields, personal data export, and account deletion features. Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010) and Republic Act No. 10639 (Free Mobile Disaster Alerts Act) establish the importance of timely public advisories.

Related Studies and Systems

Foreign Systems

TripAdvisor and Google Maps. These global platforms provide listings, photos, maps, directions, and user reviews for attractions and establishments worldwide. Their strengths are reach and volume of content. However, their information is not maintained by the local government, coverage of small local businesses is uneven, and they do not deliver official municipal advisories or give the local tourism office moderation control or analytics.

Facebook Pages. Many local tourism offices, including Daet's, use Facebook pages to announce events and answer questions. These pages are free and familiar, but posts are not structured as searchable listings, older information is buried quickly, inquiries are difficult to track, and the page owner has no access to structured analytics on the places tourists are interested in.

Local Systems and Studies

Travel Philippines (Department of Tourism). [Verify and describe the Department of Tourism's official Travel Philippines application: its features, coverage, and whether it includes municipal-level details and real-time local advisories. Cite the official source.]

Emergency Cell Broadcast System. [Describe the national mobile alert system implemented under RA 10639 and how its scope (national/regional disaster alerts via mobile networks) differs from tourism-specific municipal advisories. Cite the official source.]

Viajefy: A Tourism Information System using TF-IDF Algorithm (Arruejo & Arruejo, 2025). Researchers from the University of Northern Philippines developed and evaluated Viajefy, a tourism information system for Ilocos Sur that uses the Term Frequency–Inverse Document Frequency (TF-IDF) algorithm as a recommender to help the provincial government promote attractions and establishments. The system received a "Very Highly Acceptable" rating with a mean of 4.74. Like Connect Daet, it is a government-oriented tourism portal evaluated through a Likert-scale questionnaire; unlike Connect Daet, it focuses on recommendations and does not include real-time alerts, moderated community features, or role-based staff administration.

Local Study 2. [Insert a 2022–2026 Philippine study on an LGU web portal, e-governance service, or SMS/web alert system: author, year, title, findings, and limitations.]

Local Study 3. [Insert another 2022–2026 Philippine study that evaluated a system using ISO/IEC 25010: author, year, title, and evaluation results.]

Synthesis

The reviewed literature establishes that destinations need accurate, timely, and official online information (Buhalis & Law, 2008; Gretzel et al., 2015), that user-generated content on general platforms is influential but uncontrolled (Xiang & Gretzel, 2010), and that web technologies such as PWAs, WebSockets, and Web Push make app-like, real-time delivery possible without native applications. Security and quality frameworks (OWASP, RBAC, ISO/IEC 25010:2023) define how such a system should be built and evaluated.

The reviewed systems each cover part of the need. Recent local work such as Viajefy (Arruejo & Arruejo, 2025) shows that provincial tourism portals are acceptable to users, but it focuses on recommendations rather than real-time advisories or moderated engagement. Global platforms offer listings and reviews but no local government control or official advisories; social media offers reach but no structure, moderation workflow, or analytics; and national systems offer advisories or national information but not municipal tourism detail. Connect Daet is unique in combining, in one locally managed platform: (1) a structured directory of attractions, accommodations, restaurants, and businesses with maps and directions; (2) category-targeted real-time alerts with scheduling, "all clear" follow-ups, and delivery and reach tracking; (3) pre-moderated community features with reports, warnings, bans, suspensions, and appeals; (4) official replies to reviews and tracked inquiries; and (5) role-based administration with audit logs and exportable analytics.

Feature	Google Maps / TripAdvisor	Facebook Page	National apps	Connect Daet
Official, LGU-maintained listings	No	Partial	Partial	Yes
Search and filters by type, category, barangay, rating	Yes	No	Partial	Yes
Category-targeted real-time municipal alerts	No	No	Partial	Yes
Scheduled alerts with delivery and reach counts	No	No	No	Yes
Moderated blogs and forums	No	No	No	Yes
Official replies to reviews	Partial	Partial	No	Yes

Feature	Google Maps / TripAdvisor	Facebook Page	National apps	Connect Daet
Tracked inquiries with status	No	No	No	Yes
Role-based admin console and audit logs	No	No	N/A	Yes
Analytics export (PDF/Excel)	No	Partial	N/A	Yes

Table 1. Comparison of Existing Systems and Connect Daet [update after verifying each local system]

Theoretical Framework

This study is anchored on two established models.

Technology Acceptance Model (TAM). Davis (1989) proposed that a user's intention to use a technology is determined mainly by its perceived usefulness (the degree to which it improves performance) and perceived ease of use (the degree to which it is free of effort). For Connect Daet, perceived usefulness comes from accurate listings, timely alerts, and tracked inquiries, while perceived ease of use comes from a mobile-first interface, clear navigation, and minimal steps. This model justifies the study's priority on usability.

DeLone and McLean Information Systems Success Model. DeLone and McLean (2003) identified information quality, system quality, and service quality as the drivers of use and user satisfaction, which in turn produce net benefits for individuals and organizations. In this study, information quality is ensured by LGU-maintained content and moderation, system quality by security and real-time performance, and service quality by inquiry handling and official replies.

ISO/IEC 25010:2023 Product Quality Model. Of the nine quality characteristics in ISO/IEC 25010:2023, this study evaluates **Usability**, termed **Interaction Capability** in the 2023 revision. Its sub-characteristics are appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness. Usability is the measurable counterpart of perceived ease of use in TAM, which is why it was chosen as the focus of the evaluation.

[Insert diagram linking TAM (perceived usefulness, perceived ease of use) and the DeLone and McLean model (information, system, and service quality → use and satisfaction → net benefits), with ISO/IEC 25010 as the measure of system quality]

Figure 2. Theoretical Paradigm

Conceptual Framework

The study follows the Input-Process-Output (IPO) model shown in Figure 3.

Input	Process	Output
Knowledge requirements: Tourism information practices of the Daet Municipal Tourism Office; problems in current dissemination and engagement; ISO/IEC 25010:2023; OWASP Top 10; RA 10173. Software requirements: Next.js, React, Tailwind CSS; Supabase (PostgreSQL, Auth, Storage, Realtime, Edge Functions); Web Push (VAPID); SendGrid; Google Maps; OpenWeather; Git and Visual Studio Code. Hardware requirements: Development laptops; Android and iOS phones and tablets for testing; internet connection.	Agile Scrum development: Requirements gathering (interviews, observation); sprint planning; design (use case, DFD, ERD, UI); development in sprints (accounts and roles; places and events; alerts and push; reviews, blogs, forums; moderation; analytics; settings, audit, backup); testing (unit, black-box, security); deployment. Evaluation: ISO/IEC 25010:2023 Usability questionnaire (five-point Likert scale) answered by tourists, tourism office staff, and IT experts; statistical treatment using weighted mean.	Daet Tourist Information, Engagement, and Management System with Real-Time Updates (Connect Daet) A secure, mobile-responsive PWA with a public places and events portal, real-time alerts, moderated community features, and a role-based administrative console with analytics. Evaluation results on the system's acceptability.

[Insert IPO diagram with feedback arrow from Output back to Process]

Figure 3. Conceptual Paradigm

CHAPTER 3

METHODOLOGY

This chapter presents the operational framework and engineering approach used to develop, implement, and validate the study. It details the procedures used to translate requirements into a working system and to evaluate its quality.

Research Design

This study uses a **descriptive-developmental research design**. The developmental part covers the analysis, design, construction, and testing of Connect Daet using an established software development life cycle. The descriptive part gathers and describes the current tourism information practices and problems of the Daet Municipal Tourism Office through interviews and observation, and later describes the respondents' evaluation of the developed system using a standardized questionnaire based on ISO/IEC 25010. This combination is appropriate because the study must both produce a working system and provide measurable evidence of its quality.

Participants of the Study

The respondents are divided into three groups:

Group	Description	Sampling technique	Number
Tourism office personnel	Municipal Tourism Officer and staff who will use the administrative console	Total enumeration / purposive	[n]
IT experts	IT professionals or faculty with at least [2] years of experience in web development or systems administration	Purposive	[n, e.g., 5]
Tourists and residents	Visitors and residents of Daet aged 18 and above who use a smartphone or computer	Slovin's formula, convenience sampling	[n]
Total			[N]

Table 2. Distribution of Respondents

The sample size for tourists and residents is computed using Slovin's formula (Tejada & Punzalan, 2012):

$$n = N / (1 + N e^2)$$

where n is the sample size, N is the population size [insert the population used, e.g., average monthly tourist arrivals in Daet, with source], and e is the margin of error (0.05). [Show the computation and final sample size.] Inclusion criteria: respondents must be at least 18 years old, have used the system during the testing period, and consent to participate. Respondents who did not complete the questionnaire are excluded.

Data Gathering Techniques

Interview. Semi-structured interviews with the Municipal Tourism Officer and staff were conducted to identify current practices, problems, and requirements (SOP 1 and SOP 2). [Insert interview dates and attach the interview guide in the appendix.]

Observation and document analysis. The researchers reviewed the office's Facebook page, existing printed and online materials, and inquiry records to document information gaps.

Questionnaire. A survey questionnaire based on the Usability (Interaction Capability) characteristic of ISO/IEC 25010:2023 was used to evaluate the system (SOP 4). It contains statements grouped under the eight usability sub-characteristics (appropriateness recognizability, learnability, operability, user error protection, user engagement, inclusivity, user assistance, and self-descriptiveness), each rated on a five-point Likert scale (Likert, 1932) from 5 (Strongly Agree) to 1 (Strongly Disagree). The instrument was validated by [number] experts and pilot-tested with [number] respondents before deployment. [Attach the questionnaire in the appendix.]

System testing. Unit tests, black-box functional tests, role-permission tests, and security checks were performed during development (see Chapter 4, Development and Testing).

Scale	Range	Verbal interpretation
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Acceptable

Scale	Range	Verbal interpretation
3	2.61 – 3.40	Moderately Acceptable
2	1.81 – 2.60	Slightly Acceptable
1	1.00 – 1.80	Not Acceptable

Table 3. Likert scale and verbal interpretation

Sources of Data

Primary sources. Interview responses of the tourism office personnel; questionnaire responses of tourism office personnel, IT experts, and tourists; and test results recorded by the researchers.

Secondary sources. Books, journal articles, standards (ISO/IEC 25010:2023, OWASP Top 10, WCAG 2.1), Philippine laws (RA 9593, RA 7160, RA 10173, RA 10121, RA 10639), Department of Tourism and Philippine Statistics Authority data for 2022–2025, tourist arrival records of the Daet Municipal Tourism Office, and technical documentation of the frameworks and services used (Next.js, Supabase, Web Push).

Statistical Treatment of Data

Frequency and percentage. Used to describe the profile of respondents.

$$P = (f / n) \times 100$$

where P is the percentage, f is the frequency, and n is the total number of respondents.

Weighted mean. Used to determine the respondents' rating of each usability statement, each sub-characteristic, and Usability as a whole.

$$WM = \Sigma(f \times w) / n$$

where WM is the weighted mean, f is the frequency of each response, w is the weight of the response (1 to 5), and n is the number of respondents. The result is interpreted using Table 3. Data were tabulated and computed using [Microsoft Excel / SPSS].

Software Development Methodology

The system was developed using **Agile Scrum** (Schwaber & Sutherland, 2020). Scrum organizes work into short, fixed-length sprints, each producing a working increment

that is reviewed with stakeholders. It was chosen because requirements from the tourism office were refined over time, and each module could be demonstrated and adjusted before the next was built. The development team assigned the roles of Product Owner ([name]), Scrum Master ([name]), and Development Team ([names]).

[Insert Scrum cycle diagram: product backlog → sprint planning → sprint (daily scrum) → sprint review → retrospective → increment]

Figure 4. Agile Scrum Model

Requirements Gathering and Planning

Interviews and observations identified the problems described in Chapter 1. Requirements were written as a product backlog grouped by user role (Visitor, Registered Tourist, Tourism Officer, Moderator, Administrator) and prioritized with the tourism office.

Sprints

The backlog was delivered in seven sprints:

1. **Sprint 1 – Accounts and roles:** database schema, registration, sign-in, password recovery, and role-based access control for five roles.
2. **Sprint 2 – Places directory and events:** attractions, accommodations, restaurants, and businesses with photos, maps, hours, fees, categories, tags, featured status, and archiving; events with approval; public pages and search.
3. **Sprint 3 – Real-time alerts:** alert categories, composer, scheduling, targeting, templates, "all clear," delivery and reach tracking, and Web Push.
4. **Sprint 4 – Community features:** reviews and ratings, blogs, forums, reactions, favorites, and inquiries.
5. **Sprint 5 – Moderation:** pre-moderation queues, reports, profanity filter, warnings, posting bans, suspensions, appeals, and community guidelines.
6. **Sprint 6 – Analytics:** traffic, devices, popular places, ratings trends, alert reach, staff activity, and PDF/Excel export.
7. **Sprint 7 – Administration:** system settings, security policy, audit logs, failed sign-ins, backup and restore, and hardening.

Testing and Deployment

Each sprint included automated unit tests, lint and build checks, and manual testing by role. A final integration and security review was performed before deployment to the cloud host and Supabase project, followed by user evaluation.

Project Timeline

Activity	**[Month 1]**	**[Month 2]**	**[Month 3]**	**[Month 4]**	**[Month 5]**	**[Month 6]**
Requirements gathering	■					
System design	■	■				
Sprints 1–2		■				
Sprints 3–4			■			
Sprints 5–6				■		
Sprint 7 and hardening					■	
Testing and evaluation					■	■
Documentation and deployment						■

Table 4. Gantt chart [adjust months and bars to your actual schedule]

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the findings of the study according to the statement of the problem, the technical background and design of the system, its development and testing, the prototype, and the implementation plan and results.

FINDINGS OF THE STUDY

SOP 1. What are the operational challenges and communication gaps in the current tourism information dissemination and visitor engagement methods of the Municipality of Daet?

[Summarize the interview and observation results. Suggested structure: (a) single-channel dependence on the Facebook page; (b) inconsistent and outdated information; (c) untracked and repeated inquiries; (d) no official real-time advisory channel; (e) no centralized content management; (f) no structured feedback or analytics. Support each with quotes, counts, or examples gathered from the office.]

These findings confirm the need for a single, officially maintained platform with real-time alerts, tracked engagement, and management tools, which shaped the requirements in SOP 2.

SOP 2. What technical, functional, and security requirements are needed in developing a real-time tourist information and management system for the Municipality of Daet?

The requirements identified are summarized below.

Category	Requirement
Functional – Visitors	Browse and search places and events; filter by type, category, barangay, and rating; view photos, maps, directions, hours, fees, and contacts; read blogs, forums, reviews, and alerts; read About, Contact, and Guidelines pages; subscribe a device to alerts.
Functional – Registered tourists	Register, sign in, recover password, edit profile and photo; favorites and trip planner; reviews; blogs; forum threads and replies; reactions and reports; alert category preferences; notification inbox; inquiries and feedback with status and replies.
Functional – Tourism Officers	Manage listings and events; compose, schedule, target, edit, cancel, and close alerts; templates; delivery and reach counts; reply to reviews; answer inquiries; analytics with PDF/Excel export.

Category	Requirement
Functional – Moderators	Pre-moderation queues; reports; edit, hide, delete content; profanity filter; warnings, posting bans, suspensions, appeals; forum categories, pin, lock; announcements; community guidelines; moderation log.
Functional – Administrators	Staff and tourist accounts; roles and per-person permissions; settings; security policy; audit logs and failed sign-ins; backup and restore; system overview and analytics.
Technical	Mobile-responsive PWA; real-time page updates and Web Push; cloud database with storage; PDF/Excel generation; scheduled jobs.
Security	Hashed passwords (Supabase Auth); signed HTTP-only sessions with lifetime and idle timeout; server-side RBAC on every admin API; row-level security; login lockout; rate limits; input sanitization; upload content checks; private account columns hidden; audit logging; RA 10173 compliance.

Table 5. System requirements

SOP 3. How can the system be designed and developed to provide a centralized mobile-responsive portal, real-time alerts, interactive features, and an administrative console with role-based access and analytics?

The system was designed and developed as described in the succeeding sections: Technical Background, Design, Software and Systems Architecture, Development and Testing, and Description of the Prototype. Each of the four required capabilities is implemented as follows: (a) the places directory and event pages under /info/visitor; (b) the alerts module with in-app and push delivery; (c) reviews, blogs, forums, and inquiries with pre-moderation; and (d) the role-based console under /info/admin with analytics.

SOP 4. How acceptable is the developed system in terms of Usability based on the ISO/IEC 25010:2023 software quality model?

[Present the respondents' profile (frequency and percentage), then one table per characteristic, then a summary table. Discuss each result.]

[For each sub-characteristic, present a table of its individual statements with their weighted means, then discuss.]

Usability sub-characteristic	Tourists	Staff	IT Experts	Overall WM	Interpretation
Appropriateness recognizability	[ ]	[ ]	[ ]	[ ]	[ ]
Learnability	[ ]	[ ]	[ ]	[ ]	[ ]

Usability sub-characteristic	Tourists	Staff	IT Experts	Overall WM	Interpretation
Operability	[ ]	[ ]	[ ]	[ ]	[ ]
User error protection	[ ]	[ ]	[ ]	[ ]	[ ]
User engagement	[ ]	[ ]	[ ]	[ ]	[ ]
Inclusivity	[ ]	[ ]	[ ]	[ ]	[ ]
User assistance	[ ]	[ ]	[ ]	[ ]	[ ]
Self-descriptiveness	[ ]	[ ]	[ ]	[ ]	[ ]
Overall Usability	[ ]	[ ]	[ ]	[ ]	[ ]

Table 6. Summary of the Usability Evaluation

TECHNICAL BACKGROUND OF THE PROJECT

The system was built with widely used, well-documented open technologies so that it can be maintained by the LGU or future developers.

Layer	Technology	Reason for choice
Front end	Next.js 16 (App Router), React 19, Tailwind CSS 4, Lucide icons	Component-based UI, server and client rendering, responsive utility styling.
Server / API	Next.js Route Handlers (Node.js runtime), Next.js Proxy	API endpoints and request-level session, role, idle-timeout, and maintenance checks in the same codebase.
Database	PostgreSQL via Supabase (104 migrations, 81 tables)	Relational integrity, row-level security, functions and triggers, managed backups.
Authentication	Supabase Auth + signed HTTP-only session cookie	Hashed passwords, email verification, recovery; tamper-proof role in session.
Real time	Supabase Realtime (WebSocket), Web Push (VAPID) via Supabase Edge Function, pg_cron	Live page refresh, device notifications, scheduled alert delivery.
Storage	Supabase Storage	Photos and videos for listings, blogs, and profiles.
Integrations	SendGrid (email), Google Maps (maps and directions), OpenWeather (dashboard weather)	Proven third-party services.
Reports	pdfkit, ExcelJS	PDF and Excel analytics and feedback exports.
Hosting	Vercel (with scheduled cron jobs) [confirm]	Serverless hosting with HTTPS.
Tools	Visual Studio Code, Git, ESLint, Node test runner	Development, version control, quality checks.

Table 7. Development tools and technologies

Design of Software, System, Product, or Processes

A. Conceptual Design of the Proposed System

The system has five actors and the following major modules: Places Directory, Events, Alerts, Community (reviews, blogs, forums), Feedback and Inquiries, Notifications, Moderation, Analytics, Accounts and Roles, and Settings and Security.

[Insert context diagram: Connect Daet at the center; Visitor, Registered Tourist, Tourism Officer, Moderator, Administrator; external services Supabase, Web Push service, SendGrid, Google Maps, OpenWeather]

Figure 5. Context Diagram

[Insert use case diagram of the five actors and their use cases from Table 5]

Figure 6. Proposed System Use Case

B. Logical Design of the Proposed System

The logical design describes the data flows and processes independent of the platform.

[Insert DFD showing processes: 1.0 Manage Accounts, 2.0 Manage Places and Events, 3.0 Broadcast Alerts, 4.0 Community Engagement, 5.0 Moderate Content, 6.0 Handle Inquiries, 7.0 Generate Analytics; data stores for users, listings, events, alerts, notifications, content, reports, audit log]

Figure 7. Data Flow Diagram (Level 0 and Level 1)

[Insert ERD of the core entities in Table 13]

Figure 8. Entity-Relationship Diagram

[Insert activity diagram: officer composes → chooses category, severity, audience → send now or schedule → system writes inbox rows → requests push → records counts → officer may edit, cancel, or mark all clear]

Figure 9. Activity Diagram – Sending an Alert

[Insert activity diagram: tourist submits → status pending → moderator approves or rejects → author notified → published content visible]

Figure 10. Activity Diagram – Pre-moderated Post

C. Physical Design of the Proposed System

The physical design is presented through the database design, the architecture, network layout, and screen designs in the sections below.

Software Architecture

c.1. Database Design

The database is a PostgreSQL schema managed through versioned SQL migrations. It contains 81 tables. Row-level security is enabled on the tables, and privileged operations run through server routes or database functions. The core tables are:

Table	Function
info_users, profiles	Accounts, roles (tourist, business, tourism_officer, moderator, admin), status, and public profile data. Private columns (email, phone, address) are not readable through the public API.
staff_permission_overrides, user_session_versions	Per-person permission adjustments; session revocation on role or password change.
info_tourist_spots, listing_categories	Places directory (attractions, accommodations, restaurants, businesses) and managed categories.
info_events, event_registrations	Events and festivals with approval workflow; attendee registrations.
alerts, alert_categories, alert_templates, alert_preferences, push_subscriptions	Alert broadcasts with schedule, status, delivery and push counts; categories; templates; subscriber preferences and devices.
info_notifications	Notification inbox for alerts, replies, approvals, and warnings.
attraction_ratings, review_helpfulness	Reviews and ratings with official office replies; helpful votes.
info_blogs, forum_threads, forum_replies, forum_categories, content_comments, content_reactions	Community content and interactions.
info_moderation, admin_user_bans, user_warnings, user_moderation_appeals	Reports, posting bans and suspensions, warnings, and appeals.
info_inquiries, info_feedback	Visitor inquiries and feedback with status and replies.
user_favorites	Saved items and trip plan dates and notes.
user_points, user_badges, reward_history	Gamification.
site_page_views	Consent-based, anonymized visit analytics.
info_audit_log, auth_login_failures, user_activity_log	Audit trail, lockout tracking, sign-in history.
admin_app_settings, admin_backup_settings	System settings and scheduled backups.

Table 8. Core database tables [attach the full data dictionary in the appendix]

c.2. Software Design Pattern

- **Client–Server with a Backend-for-Frontend API.** React pages call Next.js Route Handlers, which validate input and authorization before reading or writing the database with a server-only key.
- **Proxy / Gatekeeper.** A request proxy verifies the signed session, its version, idle timeout, page permissions, and maintenance mode before any protected page loads.
- **Policy (single rules file).** One RBAC rules module defines roles, permissions, page rules, and API rules, used by the proxy, the APIs, and the menus.
- **Publish–Subscribe (Observer).** Database changes are published through Supabase Realtime; subscribed pages refresh. Alerts fan out to subscribers through one database function and one push request.
- **State machine.** Alerts move through draft → scheduled → sent → (all clear) or cancelled; community posts through pending → published or rejected.

c.3. User Interface Design

The interface is mobile-first. It uses a blue primary color for main actions, green for success, yellow/amber for warnings, and red for errors and critical alerts, on white and light-gray surfaces with rounded cards. Buttons and tap targets are at least 44 pixels high. Public pages use a top header and a bottom navigation bar on phones (Home, Explore, Map, Profile); the administrative console uses a collapsible left sidebar. The system supports light and dark themes, reduced motion, and English, Filipino, and Bikol interface labels.

[Insert wireframe of public home and place detail]

Figure 11. Layout of the public portal (mobile and desktop)

[Insert wireframe of console with sidebar and dashboard cards]

Figure 12. Layout of the administrative console

Systems Architecture

c.4. Architecture Design

Users' browsers connect over HTTPS to the Next.js application on the cloud host. The application serves pages and API routes; API routes connect to Supabase for the

database, authentication, and storage. Browsers also hold a WebSocket connection to Supabase Realtime. When an alert is sent, a database function writes the inbox rows and calls a Supabase Edge Function secured by a shared secret, which sends push messages through the browsers' push services. Scheduled jobs (pg_cron and cloud cron) deliver scheduled alerts, run backups, and process account deletions.

[Insert architecture diagram: Browser/PWA ↔ Next.js (pages, API, proxy) ↔ Supabase (PostgreSQL, Auth, Storage, Realtime, Edge Function) ↔ Web Push services; SendGrid; Google Maps; OpenWeather]

Figure 13. System Architecture

c.5. Hardware Infrastructure

No on-premise server is required. The application and database run on cloud infrastructure. The tourism office needs only computers or phones with a browser and internet access.

c.6. Network Connectivity Design

All traffic uses HTTPS (TLS). Office devices connect through the office network or mobile data to the internet; visitors connect through mobile data or Wi-Fi. Real-time updates use secure WebSockets (WSS).

[Insert network diagram: office PCs and visitor phones → router/ISP or mobile network → internet → cloud host and Supabase]

Figure 14. Network Diagram

c.7. Minimum Hardware, Software, and Network Requirements

Item	Minimum	Recommended
Computer (office)	Dual-core CPU, 4 GB RAM	Quad-core CPU, 8 GB RAM
Mobile device	Android 8 or iOS 16.4 (for push) with 2 GB RAM	Android 11+ or iOS 17+, 4 GB RAM
Browser	Current Chrome, Edge, Firefox, or Safari	Latest Chrome or Safari
Internet	5 Mbps	20 Mbps or higher
Hosting / services	Supabase and cloud host free tiers	Paid tiers for backups, bandwidth, and uptime

Table 9. Minimum and recommended requirements [confirm against your test devices]

Development and Testing

Development followed the seven sprints described in Chapter 3. The following verification methods were used:

- **Unit and rule tests.** 160 automated tests (Node.js test runner) cover permission rules for each role, settings validation, password policy, session idle timeout, upload content checks, error-message sanitization, alert input rules, analytics exports, and migration safety rules. All tests passed.
- **Static analysis and build.** ESLint reported no errors, and the production build compiled successfully.
- **Role access tests.** Every administrative API returned 401 without a session, and role rules deny pages and APIs outside each role's permissions.
- **Security review.** Checks based on the OWASP Top 10 led to: server-side authorization on all admin APIs; login lockout shared by sign-in, account recovery, password change, and deletion; rate limits on registration, verification emails, reports, and guest alert sign-up; restriction of notification links to the site; sanitization of blog HTML; file-signature checks on uploads; hiding of private account columns; and replacement of raw database errors with safe messages.
- **Black-box tests.** [Insert the test case table: ID, module, steps, expected result, actual result, pass/fail, for each role.]

ID	Test case	Expected result	Result
TC-01	Visitor filters restaurants by barangay	Only matching listings are shown	[ ]
TC-02	Officer schedules an alert for the weather category	Subscribers of that category receive it at the scheduled time	[ ]
TC-03	Tourist submits a forum post	Post is pending until a moderator approves it	[ ]
TC-04	Moderator suspends a user for 3 days	User is signed out and cannot sign in until expiry	[ ]
TC-05	Five wrong passwords	Account sign-in is locked for the configured duration	[ ]
TC-06	Tourist opens an admin URL	Access denied	[ ]

ID	Test case	Expected result	Result
TC-07	Officer exports analytics to Excel	File downloads with traffic, ratings, and alert reach sheets	[ ]

Table 10. Sample black-box test cases

Description of the Prototype

[For each screen, insert a screenshot and a short description of what the user sees and can do.] Recommended screens:

- Visitor home page and places directory with tabs (See & Do, Stay, Eat & Drink, Shops & Services) and filters.
- Place detail page with photos, map, directions, hours, fees, reviews, and office replies.
- Public alerts page and an alert detail.
- Registration, sign-in, and password recovery.
- Registered tourist dashboard, saved items and trip planner, notification inbox, and feedback page.
- Blog and forum pages.
- Tourism Officer dashboard, places management, alert composer and history, reviews page, and analytics with export.
- Moderator dashboard, moderation queues, restrict dialog, and community guidelines editor.
- Administrator overview, users and permissions, settings (general, categories, security, notifications, email), failed sign-ins, audit logs, and backups.

Implementation Plan

1. **Environment setup:** create the production Supabase project and cloud host; configure environment variables (Supabase keys, session secret, SendGrid, VAPID keys, OpenWeather, cron secret).
2. **Database migration:** apply all migrations in order; deploy the push Edge Function and store its URL and secret in Supabase Vault; verify the scheduled jobs.

3. **Data migration:** encode existing attractions, establishments, and events from the office's records and Facebook page into the directory; upload photos.
4. **Accounts:** create administrator, Tourism Officer, and Moderator accounts; configure settings, categories, and community guidelines.
5. **Training:** conduct hands-on sessions for each staff role, with a user manual. [schedule]
6. **Pilot and launch:** pilot with selected users for [period], then announce the official portal through the office's Facebook page.
7. **Maintenance:** monitor audit logs and backups; apply updates.

Activity	Responsible	Target date
Environment and database setup	Developers	[ ]
Content encoding	Tourism office staff with developers	[ ]
Training	Developers	[ ]
Pilot testing	Tourism office and selected users	[ ]
Official launch	Daet Municipal Tourism Office	[ ]

Table 11. Implementation schedule

Implementation Results

[Present the results of the pilot deployment and the Usability evaluation (summarize the Summary of the Usability Evaluation table), plus qualitative feedback from tourism office staff and users, and any measurable operational changes observed (e.g., inquiries answered, alerts sent, reach).]

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the findings discussed in the previous chapter, the conclusions drawn, and the recommendations of the study.

Summary of Findings

SOP 1. [In the past tense, summarize the challenges found, e.g., "The Daet Municipal Tourism Office relied solely on its Facebook page… ," with any counts gathered.]

SOP 2. The study identified functional requirements for five user roles, technical requirements for a mobile-responsive PWA with real-time updates and push notifications, and security requirements including server-side role-based access control, row-level security, login lockout, rate limiting, input sanitization, upload checks, and audit logging.

SOP 3. The system was developed in seven Agile sprints using Next.js and Supabase, producing a places and events portal, a real-time alert module with scheduling, targeting, and reach tracking, pre-moderated community features, and a role-based administrative console with analytics exports. It passed all 160 automated tests, compiled without errors, and denied unauthorized access to all administrative APIs.

SOP 4. [State the overall weighted mean of Usability and of each sub-characteristic with its verbal interpretation, e.g., "Usability obtained an overall weighted mean of 4.52, interpreted as Highly Acceptable; learnability obtained the highest mean…"]

Conclusions

SOP 1. Based on the findings, it is concluded that the tourism office's dependence on a single social media channel causes fragmented information, untracked inquiries, and the absence of official real-time advisories.

SOP 2. It is concluded that a real-time tourism system for a municipality must combine role-specific functions with layered security enforced on the server and in the database, not only in the user interface.

SOP 3. It is concluded that a single locally managed web platform can centralize tourism information, deliver targeted real-time alerts, and support moderated engagement and data-driven management.

SOP 4. [Write a categorical, number-free conclusion, e.g., "The developed system is highly usable for its intended users, particularly tourists on mobile devices."]

Recommendations

1. The **Daet Municipal Tourism Office** should adopt Connect Daet as its official tourism portal, assign staff to keep listings, events, and alerts current, and promote the portal through its Facebook page and printed materials such as QR codes at attractions.
2. The **LGU of Daet** should fund paid cloud hosting and database plans to ensure backups, bandwidth, and uptime, and register an official domain name.
3. The **system administrator** must apply all database migrations, keep secrets and keys rotated, review audit logs and failed sign-ins regularly, and clear legacy credential data from older records.
4. The **Moderators** should review the community guidelines and the profanity word list periodically with the tourism office.
5. The **Municipal Disaster Risk Reduction and Management Office** may coordinate with the tourism office so that urgent advisories are issued through the alert module alongside official emergency channels.
6. **Future researchers** may add SMS delivery for alerts to reach users without smartphones, publish native mobile applications, add online booking and payment for accredited establishments, provide automatic translation of user content, extend coverage to all municipalities of Camarines Norte, and measure the system's long-term effect on tourist arrivals and satisfaction.

REFERENCES

A. Published Materials

Arruejo, A. C., & Arruejo, R. C. (2025). Development and evaluation of Viajefy: A tourism information system using TF-IDF algorithm. *Mindanao Journal of Science and Technology, 23*(1), 80–101. https://doi.org/10.61310/mjst.v23i1.2383

Biørn-Hansen, A., Majchrzak, T. A., & Grønli, T.-M. (2017). Progressive web apps: The possible web-native unifier for mobile development. In *Proceedings of the 13th International Conference on Web Information Systems and Technologies (WEBIST 2017)* (pp. 344–351). SciTePress.

Buhalis, D., & Law, R. (2008). Progress in information technology and tourism management: 20 years on and 10 years after the Internet—The state of eTourism research. *Tourism Management, 29*(4), 609–623.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly, 13*(3), 319–340.

DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems, 19*(4), 9–30.

Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From game design elements to gamefulness: Defining "gamification." In *Proceedings of the 15th International Academic MindTrek Conference* (pp. 9–15). ACM.

El Archi, Y., Benbba, B., Nizamatdinova, Z., Issakov, Y., Vargáné, G. I., & Dávid, L. D. (2023). Systematic literature review analysing smart tourism destinations in context of sustainable development: Current applications and future directions. *Sustainability, 15*(6), 5086. https://doi.org/10.3390/su15065086

Gretzel, U., Sigala, M., Xiang, Z., & Koo, C. (2015). Smart tourism: Foundations and developments. *Electronic Markets, 25*(3), 179–188.

Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology, 22*(140), 1–55.

Sandhu, R. S., Coyne, E. J., Feinstein, H. L., & Youman, C. E. (1996). Role-based access control models. *Computer, 29*(2), 38–47.

Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson.

Tejada, J. J., & Punzalan, J. R. B. (2012). On the misuse of Slovin's formula. *The Philippine Statistician, 61*(1), 129–136.

Xiang, Z., & Gretzel, U. (2010). Role of social media in online travel information search. *Tourism Management, 31*(2), 179–188.

B. Standards, Reports, and Online Sources

Bicol Mail. (2025). *Bicol tourism booms with P3.5-B in domestic receipts*. https://www.bicolmail.net/single-post/bicol-tourism-booms-with-p3-5-b-in-domestic-receipts

Department of Tourism. (2023). *National Tourism Development Plan 2023–2028*. Republic of the Philippines.

DZRH News. (2026). *PH logs 6.48M international visitors, still short of pre-pandemic tourism levels — DOT*. https://www.dzrh.com.ph/post/ph-logs-648m-international-visitors-still-short-of-pre-pandemic-tourism-levels-dot

Fette, I., & Melnikov, A. (2011). *The WebSocket protocol* (RFC 6455). Internet Engineering Task Force.

International Organization for Standardization. (2023). *ISO/IEC 25010:2023 Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model*.

Manila Bulletin. (2025, January 21). *DOT records 5.9M foreign tourist arrivals in 2024 but misses target due to decline in Chinese visitors*. https://mb.com.ph/2025/1/21/dot-records-5-9-m-foreign-tourist-arrivals-in-2024-but-misses-target-due-to-decline-in-chinese-visitors

OWASP Foundation. (2021). *OWASP Top 10:2021*. https://owasp.org/Top10/

Philippine Statistics Authority. (2024). *Camarines Norte's economy records 3.7 percent increase in 2023*. https://rsso05.psa.gov.ph/content/camarines-nortes-economy-records-37-percent-increase-2023

Schwaber, K., & Sutherland, J. (2020). *The Scrum guide*. https://scrumguides.org

Thomson, M., Damaggio, E., & Raymor, B. (2016). *Generic event delivery using HTTP push* (RFC 8030). Internet Engineering Task Force.

World Wide Web Consortium. (2018). *Web Content Accessibility Guidelines (WCAG) 2.1*. https://www.w3.org/TR/WCAG21/

C. Laws

Republic Act No. 7160. (1991). *Local Government Code of 1991*.

Republic Act No. 9593. (2009). *Tourism Act of 2009*.

Republic Act No. 10121. (2010). *Philippine Disaster Risk Reduction and Management Act of 2010*.

Republic Act No. 10173. (2012). *Data Privacy Act of 2012*.

Republic Act No. 10639. (2014). *Free Mobile Disaster Alerts Act*.

[Add the remaining local studies and the Travel Philippines/ECBS sources you cite in Chapter 2. Verify every entry and access date against the original source.]

APPENDICES

APPENDIX A

INTERVIEW GUIDE AND TRANSCRIPTS

[Insert interview guide and transcripts — rename this appendix if your school requires different contents.]

APPENDIX B

USABILITY SURVEY QUESTIONNAIRE (ISO/IEC 25010:2023)

[Insert usability survey questionnaire (iso/iec 25010:2023) — rename this appendix if your school requires different contents.]

APPENDIX C

DATA DICTIONARY

[Insert data dictionary — rename this appendix if your school requires different contents.]

APPENDIX D

USER MANUAL

[Insert user manual — rename this appendix if your school requires different contents.]

CURRICULUM VITAE

[Insert each proponent's curriculum vitae.]